INCH-POUND

MIL-PRF-32239
3 May 2007

PERFORMANCE SPECIFICATION

COATING SYSTEM, ADVANCED PERFORMANCE, FOR AEROSPACE APPLICATIONS

This specification is approved for use by all Departments and Agencies of the Department of Defense

1. SCOPE

1.1 Form.

This specification establishes the requirements for coating systems having high durability to be used in finishing aircraft exterior surfaces. These coating systems normally are comprised of a surface treatment, a primer, and a topcoat and include a chemical stripper and a cleaning compound for topcoat washing.

1.2 Application.

Systems are qualified for use on external surfaces of military aircraft but are not limited to such application. If the coating systems qualified to this specification are to be used on other than external surfaces, further testing will be required for the specific application.

1.3 Classification.

The systems under this specification consist of the following types, classes, and grades.

Type 1: Coating system containing chromium compounds.

Type 2: Coating system, chromium free.

Class 1: Standard Flexibility Coating System

Class 2: High Flexibility Coating System

Grade 1: Standard Pigmented Coating System

Grade 2: Low IR Pigmented Coating System

Comments, suggestions, or questions on this document should be addressed to ASC/ENOI, 2530 Loop Road West, Wright-Patterson AFB OH 45433-7101 or emailed to Engineering.Standards@wpafb.af.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at http://assist.daps.dla.mil.

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1.4 Colors.

Coating systems should be furnished in all colors and gloss ranges as designated in the current issue of FED-STD-595. Colors not in FED-STD-595 should be furnished as specified by the procuring activity.

1.5 Safety-hazardous materials.

While the materials, methods, applications, and processes described or referenced in this specification may involve the use of hazardous materials, this specification does not address the hazards which may be involved in such use. It is the sole responsibility of the user to ensure familiarity with the safe, proper use and disposal of any hazardous materials and to take necessary precautionary measures to ensure the health and safety of all personnel involved.

2. APPLICABLE DOCUMENTS

2.1 General.

The documents listed in this section are specified in sections 3, 4, or 5 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3, 4, or 5 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks.

The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

FEDERAL STANDARDS

FED-STD-141	Paint, Varnish, Lacquer and Related Materials: Methods of Inspection, Sampling and Testing
FED-STD-313	Material Safety Data, Transportation Data, and Disposal Data for Hazardous Materials Furnished to Government Activities
FED-STD-595	Colors used in Government Procurement:
FED-STD-595/36118	Gray, Lusterless (para 3.6.1)
FED-STD-595/13538	Yellow Gloss (para 3.6.5)
FED-STD-595/12197	Orange, Gloss (para 3.6.5)
FED-STD-595/11136	Red, Gloss (para 3.6.5)
FED-STD-595/17925	Miscellaneous, Gloss (para 4.5.1 e)
FED-STD-595/36173	Gray, Lusterless (para 6.3.1)

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DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-C-8507	Coating, Wash Primer (Pretreatment) for Metals, Applications of (for Aeronautical Use) *Inactive for New Design*
MIL-C-8514 (ASG)	Coating Compound, Metal Pretreatment, Resin-Acid
MIL-PRF-23699	Lubricating Oil, Aircraft Turbine Engines, Synthetic Base.
MIL-PRF-83282	Hydraulic Fluid, Fire Resistant, Synthetic Hydrocarbon Base, Aircraft.
MIL-PRF-85285	Coating: Polyurethane, Aircraft and Support Equipment
MIL-PRF-85570	Cleaning Compounds, Aircraft, Exterior
MIL-PRF-87937	Cleaning Compounds, Aerospace Equipment

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-129	Military Marking for Shipment and Storage

(Copies of these documents are available online via http://assist.daps.dla.mil/quicksearch or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.3 Other Government documents, drawings, and publications.

The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

Code of Federal Regulations

29 CFR 1910.1200	Occupational Safety and Health Standards: Material Safety Data Sheets; Preparation and Submission

(Copies are available from the Superintendent of Documents, U.S. Government Printing Office, North Capitol & “H” Streets, N.W., Washington, DC 20402-0002 or http://www.gpoaccess.gov/cfr/link.html.

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Coating Technologies Procedure

Warner-Robins ALC Chemical Stripper Purchase Description

Oklahoma City ALC Chemical Stripper Purchase Description

Chemical Strippers listed in Technical Order 1-1-8

(Copies available from AFRL/MLSSO (CTIO), Building 652 Room 122, 2179 12th Street, WPAFB, OH 45433-7718, Phone (937) 255-0954.)

2.4 Non-Government publications.

The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

SAE Publications

AMS QQ-A-250/4	Aluminum Alloy 2024, Plate and Sheet (DoD Adopted)
AMS QQ-A-250/5	Aluminum Alloy Alclad 2024, Plate and Sheet (DoD Adopted)
AMS QQ-A-250/12	Aluminum Alloy 7075, Plate and Sheet (DoD Adopted)
AMS QQ-A-250/13	Aluminum Alloy Alclad 7075, Plate and Sheet (DoD Adopted)
AMS 3819	Cloths, Cleaning, for Aircraft Primary and Secondary Structural Surfaces (DoD Adopted)
AMS 4911	Titanium Alloy, Sheet, Strip and Plate 6AL-4V, Annealed (DoD Adopted)
AS 5505	Requirements for Accreditation of Testing Laboratories for Organic Coatings (DoD Adopted)
AS 9100	Quality Management Systems - Aerospace - Requirements (DoD Adopted)

(Copies are available from SAE, 400 Commonwealth Drive, Warrendale, PA 15096-0001. Order electronic standards online at http://www.sae.org/technicalcommittees/index.htm.)

ASTM Publications

ASTM B 117	Standard Practice for Operating Salt Spray (Fog) Apparatus (DoD Adopted)

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ASTM G 154	Standard Practice for Operating Fluorescent Light Apparatus for UV Exposure of Nonmetallic Materials-Replaces ASTM G53; E1-2001
ASTM G 155	Standard Practice for Operating Xenon Arc Light Apparatus for Exposure of Non-Metallic Materials (DoD Adopted)
ASTM D 522	Standard Test Methods for Mandrel Bend Test of Attached Organic Coatings (DoD Adopted)
ASTM D 523	Standard Test Method for Specular Gloss (DoD Adopted)
ASTM D 1200	Standard Test Method for Viscosity by Ford Viscosity Cup (DoD Adopted)
ASTM D 2243	Standard Test Method for Freeze-Thaw Resistance of Water-Borne Coatings (DoD Adopted)
ASTM D 2244	Standard Practice for Calculation of Color Tolerances and Color Differences from Instrumentally Measured Color Coordinates (DoD Adopted)
ASTM D 2247	Testing Coated Metal Specimens at 100 Percent Relative Humidity. (DoD Adopted)
ASTM D 2794	Resistance of Organic Coatings to the Effects of Rapid Deformation (Impact) (DoD Adopted)
ASTM D 2803	Metal, Organic Coatings on, Filiform Corrosion Resistance of, (DoD adopted)
ASTM D 2805	Hiding Power of Paints (DoD Adopted)
ASTM D 3335	Low Concentrations of Lead, Cadmium, and Cobalt in Paint by Atomic Absorption Spectroscopy (DoD Adopted)
ASTM D 3359	Standard Test Methods for Measuring Adhesion by Tape Test

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ASTM D 3363	Standard Test Method for Film Hardness by Pencil Test (DoD Adopted)
ASTM D 3718	Standard Test Method for Low Concentrations of Chromium in Paint by Atomic Absorption Spectroscopy (DoD Adopted)
ASTM D 3960	Volatile Organic Content (VOC) of Paints and Related Coatings (DoD Adopted)

(Copies of this document are available from ASTM International, 100 Barr Harbor Drive, PO Box C700, West Conshohocken, Pennsylvania, USA 19428-2959. Order electronic standards online at www.astm.org).

ISO Publications

ISO 1513	Paints and Varnishes - Examination and Preparation of Samples for Testing
ISO 4617	Paints and Varnishes - List of Equivalent Terms
ISO 4618	Paints and Varnishes - Terms and Definitions for Coating Materials
ISO 15528	Paints, Varnishes and Raw Materials for Paints and Varnishes – Sampling

Copies are available from American National Standards Institute, Inc., 25 West 43rd Street, New York, NY 10036-8002, and International Organization for Standardization (ISO), 1, rue de Varembé, Case postale 56, CH-1211 Genève 20, Switzerland. Order electronic standards online at http://www.iso.org/iso/en/CatalogueListPage.CatalogueList.

2.5 Order of precedence.

In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 Qualification.

The coating system furnished under this specification shall be products which have been tested and have passed the qualification tests specified herein and have been listed or approved for listing on the applicable qualified products list (QPL) at the time of award of contract (see 6.2). Changes shall not be permitted in the formulation of an approved product unless specific, written approval of the qualifying activity is obtained.

3.2 Materials.

The coating system shall consist of a surface treatment, a primer, and a topcoat. The coating system shall also include, in the form of COTS products, a chemical stripper and a cleaning compound for topcoat washing. The aforementioned are the minimum products necessary to achieve the performance requirements detailed herein. Unicoat systems are acceptable. Composition of the coating system components is not specified, except for the following.

3.2.1 Nonqualification of system components.

This specification does not qualify cleaners or chemical strippers. Qualified products for cleaners shall be selected from the Qualified Products List (QPL) of MIL-PRF-87937, Type IV, or MIL-PRF-85570 Type II. Qualified products for chemical strippers shall be selected from the Warner-Robins ALC Purchase Description (U.S. Air Force), Oklahoma City ALC Purchase Description (U.S. Air Force), or listed in Technical Order 1-1-8 (U.S. Air Force).

3.3 Coating system components.

The coating system, normally comprised of a surface treatment, a primer, and a topcoat, as well as a chemical stripper and a cleaning compound for topcoat washing, shall consist of the minimum number of products necessary to achieve the performance requirements detailed herein. Unicoat systems are acceptable. Composition of the coating system components is not specified, except for the following.

3.3.1 Restrictions in chemical composition.

When tested in accordance with ASTM D 3335, no component of any system element shall contain cadmium or cadmium compounds, or more than 0.05 percent of lead or lead compounds. Type 2 system components shall not contain chromium or chromium compounds when tested in accordance with ASTM D 3718.

3.3.2 Toxicity.

Manufacturers of system components shall certify that any materials supplied under this specification have no adverse effect on the health of personnel when used for its intended purpose and with the precautions as identified on shipping containers and on the material safety data sheet (see 6.3.3).

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3.4 System component physical requirements.

Coating system physical requirements are summarized in Table I.

Table I. Physical requirements.

Property	Requirement	Conditioning	Test Method
Quality	3.4.1	As received in original container	ISO 1513
Storage Stability	3.4.2	As received in original container	4.6.1
Accelerated Storage Stability	3.4.3	As received in original container	4.6.2
Freeze/Thaw Stability	3.4.4	As received in original container	4.6.3
VOC content	3.5.1 Primer – 340 g/l Topcoat- 420 g/l	Mixed, prepared for application	4.6.4
Odor	3.5.2	Mixed, prepared for application	4.6.5
Viscosity/Pot Life	3.5.3	Mixed, prepared for application	4.6.6
Drying Time	3.5.4	Applied on test panels	4.6.7
Wet-Edge	3.5.5	Applied on test panels	4.6.8

3.4.1 Quality.

Coating system components taken from original containers as received shall be free from extraneous matter and show no skinning, gelling, hard settlement or other objectionable properties which will prevent satisfactory application to produce a defect free film.

All Materials or Components. Materials or components shall be capable of being reincorporated into a smooth, homogenous, and pourable mixture. It shall be free of grit, skin, seeds, lumps, foreign contaminants, and thickening or livering.

Pretreatment Powders. System components supplied in powder form shall be uniform in composition and quality, and be free from impurities and other defects that impair usability and adversely affect performance when the powder is mixed with water or other materials as specified by the manufacturer under normal agitation in preparation for use.

Mixing. All components shall be readily miscible when mixed according to manufacturer's instructions.

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3.4.2 Storage stability.

The contents of each component of the coating system or premixed liquid surface pretreatment, as packaged by the manufacturer, shall meet all requirements of this specification when tested in accordance with 4.6.1. A subset of testing shall be performed using coating system components that have been subjected to the requirements of this paragraph. The subset of requirements and corresponding tests shall be 3.5.3, 3.6.1, 3.6.4, 3.6.6.1, 3.6.7.1, 3.6.8 (MIL-PRF-83282 only).

For touch-up and repair kits, contents of the coating system material or premixed liquid surface pretreatment repackaged by manufacturers' approved packaging companies shall meet all the requirements of this specification after one year minimum indoor storage from date of repackaging, where the daily temperature of the ambient air is maintained between 35 to 95 ºF (1.7 to 35 ºC).

3.4.3 Accelerated storage stability.

Each component of the coating system excluding the system pretreatment, cleaner and chemical stripper shall meet all the requirements of this specification when tested in accordance with 4.6.2. The accelerated storage conditions shall be 7 days at 140±5 ºF (60±3 ºC)**.** A subset of testing shall be performed on the coating samples that have been subjected to the requirements of this paragraph. The subset of requirements and corresponding tests shall be 3.5.3, 3.6.1, 3.6.4, 3.6.6.1, 3.6.7.1, 3.6.8 (MIL-PRF-83282 only).

3.4.4 Freeze/thaw stability.

Water based components, excluding the system pretreatment, cleaner, and chemical stripper, as packaged by the manufacturer, shall meet all the requirements specified herein after being subjected to five cycles of freezing and thawing when tested in accordance with 4.6.3. A subset of testing shall be performed on the coating samples that have been subjected to the requirements of this paragraph. The subset of requirements and corresponding tests shall be 3.5.3, 3.6.1, 3.6.4, 3.6.6.1, 3.6.7.1, 3.6.8 (MIL-PRF-83282 only).

3.5 Admixed product properties.

3.5.1 VOC content.

The maximum VOC content at application of system coatings classified as primers shall be no greater than 340 g/l (2.8 lb/gal). The maximum VOC content at application of system coatings classified as topcoats shall be no greater than 420 g/l (3.5 lb/gal). Testing shall be in accordance with 4.6.4.

3.5.2 Odor.

The air-dried coating system shall retain no residual odor 48 hours after application when tested in accordance with 4.6.5.

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3.5.3 Viscosity and pot-life.

The viscosity/consistency of surface pretreatment materials shall be suitable for application to aircraft surfaces by spray or brush. The viscosity/consistency of primers and topcoats shall be suitable for application by conventional, HVLP, airless, and electrostatic spray (suitably equipped in the case of water based materials). Coating manufacturer shall define the recommended application viscosity range when tested in accordance with 4.6.6.

The pot-life requirement is as follows: four hours after initial mixing, primers and topcoats shall meet the performance and application requirements of this specification. A subset of testing shall be performed on the coating samples that have been subjected to the requirements of this paragraph. The subset of requirements and corresponding tests shall be 3.6.1, 3.6.4, 3.6.6.1, 3.6.7.1, 3.6.8 (MIL-PRF-83282 only). Test panels for this requirement shall be created by applying the primer 4-hours after mixing the primer, and then applying the topcoat 4-hours after mixing the topcoat.

3.5.4 Drying time.

Coating system components designated as surface pretreatments shall be dry to prime within 4 hours. Coating system components designated as primers shall be dry to topcoat within 5 hours. Coating system components designated as topcoats shall be dry to tape with no adhesion loss or permanent marring in 8 hours or less when tested in accordance with 4.6.7. As an alternative, the total time to apply pretreatment, primer, and topcoat to a dry to tape condition shall be 17 hours or less.

3.5.5 Wet-edge.

There shall be no streaks, tiger stripes, or other visual irregularities on the test panels when tested in accordance with 4.6.8.

3.6 Applied coating system requirements.

Table II lists the applied coating system requirements.

3.6.1 Surface appearance.

Using test panels created in 3.6.4, the finish shall exhibit a uniform, smooth surface, free from runs, sags, bubbles, streaks, hazing, seeding, dusting, floating, mottling, or other defects.

3.6.2 Color.

The coating system shall match the applicable color standard in FED-STD-595 with a color difference (ΔE) less than or equal to 1.0 when tested in accordance with 4.6.9.

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Table II. Cured coating system requirements.

Property	Requirement	Conditioning	Test Method
Surface Appearance	3.6.1	Air dry 24 hours minimum.	Visual Observation
Color	3.6.2 - ΔE ≤ 1		4.6.9
IR Reflectance	3.6.3 – 8% max.		4.6.10
Gloss	3.6.4 - Camouflage: 9 max. @ 85° 5 max. @ 60° Gunship: 3 max @ 85° 3 max @ 60° Semigloss: 15 to 45 max. @ 60° Gloss: 90 min. @ 60°		4.6.11
Opacity	3.6.5 – All Colors: Contrast Ratio = 0.95 Yellow (13538), Orange (12197), Red (11136): Contrast Ratio ≥ 0.90		4.6.12
Adhesion			
Wet Tape	3.6.6.1 – No delamination or peel and a rating of no less than 4A.		4.6.13
Cross-Hatch	3.6.6.2 – No delamination or peel and a rating of no less than 4B.		4.6.14
Flexibility			4.6.15
Ambient Temperature	3.6.7.1 – Class 1 Coating System ≥60% gloss, 40% camo; weathered ≥ 40% gloss, 20% camo. Class 2 Coating System ≥ 60% gloss and camo before and after weathering	Standard Cure 3.6.9 Xenon Arc 3.6.9 QUV-B	4.6.15.1
Low Temperature	3.6.7.2 – Class 1 Coating System 1” gloss, 2” camo; Class 2 Coating System 1” for gloss and camo, No Cracking	Standard Cure 3.6.9 Xenon Arc 3.6.9 QUV-B	4.6.15.2

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Table II. Cured coating system requirements – continued.

Fluid Resistance	3.6.8 – Class1 and Class 2: coating system shall show no blistering coating delamination or adhesion loss, no more than a two pencil hardness change, ΔE ≤ 3. No Skydrol reqt. for Class 2		4.6.16
MIL-PRF-23699		24 hrs @ 250 ºF	
MIL-PRF-83282		7 days @150 ºF	
Skydrol LD-4		30 days @ 77 ºF	
JP-8 +100		30 days @ 77 ºF	
DI Water		30 days @ 120 ºF	
Weathering Resistance	3.6.9 – ΔE ≤ 1 Camouflage colors: 60º 5 max. (Gunship): 60º 3 max. Semigloss colors: 60º 15 min. 45 max. Gloss colors: 60º 90 min.		4.6.17
Xenon Arc			4.6.17.1
QUV-B			4.6.17.2
Humidity Resistance	3.6.10 - Coating system: no blistering, no softening, no loss of adhesion, nor other film defects	30 days @120 ºF	4.6.18
Filiform Corrosion	3.6.11 - No filiform corrosion or blisters extending more than 1/8 inch beyond the scribe mark	Expose to HCl, 2000 hrs @ 102 ºF and 80% RH	4.6.19
Neutral Salt Spray	3.6.12 – Type 1 and Type 2: no blisters or undercutting from the scribe Type 1: no discoloration in the scribe and no pitting in the scribe Type 2: discoloration and some corrosion products allowed in the scribe but no pitting	Type 1: 3000 hrs Type 2: 2000 hrs	4.6.20
Heat Resistance	3.6.13 – no cracking, flaking or adhesion loss. ΔE ≤ 1	4 hours @ 350 ºF	4.6.21
Solvent Resistance	3.6.14 - Rubbing through to primer indicates a failure.		4.6.22
Cleanability	3.6.15 - cleaning efficiency of not less than 75%		4.6.23
Strippability	3.6.16 – 90% stripped		4.6.24

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3.6.3 Surface appearance.

Using test panels created in 3.6.4, the finish shall exhibit a uniform, smooth surface, free from runs, sags, bubbles, streaks, hazing, seeding, dusting, floating, mottling, or other defects.

3.6.4 Color.

The coating system shall match the applicable color standard in FED-STD-595 with a color difference (ΔE) less than or equal to 1.0 when tested in accordance with 4.6.9.

3.6.5 Infrared reflectance (FED-STD-595, color number 36118 only).

The total infrared reflectance (specular and diffuse) of the coating system shall be a maximum of 8 percent relative to NIST traceable Spectralon white calibration standard when tested in accordance with 4.6.10.

3.6.6 Gloss.

The specular gloss of the system, when tested in accordance with 4.6.11, shall be as follows:

Camouflage colors:	85º angle of incidence		9 max.
	60º angle of incidence		5 max.
(Gunship)	85º angle of incidence		3 max.
	60º angle of incidence		3 max.
Semigloss colors:	60º angle of incidence	15 min.	45 max.
Gloss colors:	60º angle of incidence	90 min.	

3.6.7 Opacity (hiding power).

The system topcoat, when tested in accordance with 4.6.12, shall have a contrast ratio of no less than 0.95 for all colors except yellow (FED-STD-595, Color 13538), orange (FED-STD-595, Color 12197), and red (FED-STD-595, Color 11136), which shall have a contrast ratio no less than 0.90.

3.6.8 Adhesion.

3.6.8.1 Wet tape.

The coating system shall have no less than a 4A rating and no component of the coating system shall peel or delaminate from the substrate or any other component of the system when tested in accordance with 4.6.13.

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3.6.8.2 Cross-hatch.

The coating system shall have no less than a 4B rating and no component of the coating system shall peel or delaminate from the substrate or any other component of the system when tested in accordance with 4.6.14.

3.6.9 Flexibility.

3.6.9.1 Ambient temperature impact flexibility.

Class 1 coating systems:

When tested in accordance with paragraphs 4.6.15.1, the coating system shall exhibit a minimum impact elongation of 60 percent for gloss coating systems and 40 percent for camouflage coating systems and a minimum impact elongation of 40 percent and 20 percent respectively after weathering per paragraphs 4.6.17.1 and 4.6.17.2. Flexibility testing after weathering shall not be part of the subset of tests listed in 3.4.2, 3.4.3, 3.4.4, 3.5.3.

Class 2 coating systems:

When tested in accordance with paragraphs 4.6.15.1, the coating system shall exhibit a minimum impact elongation of 60 percent for gloss and camouflage topcoats. This requirement remains the same after weathering per paragraphs 4.6.17.1 and 4.6.17.2. Flexibility testing after weathering shall not be part of the subset of tests listed in 3.4.2, 3.4.3, 3.4.4, 3.5.3.

3.6.9.2 Low temperature flexibility.

Class 1 coating systems:

The coating system shall exhibit no cracking when tested in accordance with paragraphs 4.6.15.2.

Class 2 coating systems:

The coating system shall exhibit no cracking when tested in accordance with paragraphs 4.6.15.2.

3.6.10 Fluid resistance.

The coating systems, when tested in accordance with 4.6.16 shall meet the following:

Class 1 and Class 2 Coating Systems:

The coating system shall show no blistering or coating delamination or adhesion loss when tested in accordance with 4.6.14 and shall not exhibit more than a two pencil hardness change from the original hardness when tested in accordance with 4.6.25. Skydrol resistance shall not be a requirement for Class 2 coating systems.

For all coating systems, color change (ΔE) less than or equal to 3.0 when tested in accordance with paragraph 4.6.9 is acceptable.

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3.6.11 Weathering resistance.

The coating system, when tested on separate sets of test panels (ref TABLE IV), in accordance with 4.6.17.1 or 4.6.17.2 shall

a. Have a color change (ΔE) less than or equal to 1.0 when tested in accordance with paragraph 4.6.9.

b. The 60º specular gloss of the coating system when tested in accordance with 4.6.10 shall be

Camouflage colors:	60º angle of incidence		5 max.
Gunship colors:	60º angle of incidence		3 max.
Semigloss colors:	60º angle of incidence	15 min.	45 max.
Gloss colors:	60º angle of incidence	90 min.	

c. Shall meet the requirements of 3.6.7.1 and 3.6.7.2

3.6.12 Humidity resistance.

The coating system shall not blister, nor shall it exhibit any loss of adhesion when tested in accordance with 4.6.14, nor exhibit a change in pencil hardness from the original hardness when tested in accordance with 4.6.25, nor exhibit other film defects when tested in accordance with 4.6.18.

3.6.13 Filiform corrosion resistance.

For validation of the test, the control panel shall have filiform corrosion extending a minimum of 1/4 inch beyond the scribe.

The test panels shall have no filiform corrosion or blisters extending more than 1/8 inch beyond the scribe marks upon completion of the exposure period as specified in 4.6.19

3.6.14 Neutral salt spray corrosion resistance.

The test panels shall exhibit no discoloration in the scribe, no blisters, no undercutting from the scribe and no substrate pitting in the scribe when tested in accordance with 4.6.20. Discoloration and some corrosion products in the scribe are acceptable for Type II coating systems.

3.6.15 Heat resistance.

The coating system shall have no cracking, flaking, adhesion loss when tested in accordance with 4.6.21 and a color change (ΔE) less than or equal to 1.0 when tested in accordance with paragraph 4.6.9.

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3.6.16 Solvent resistance (cure).

The coating system shall withstand 25 MEK double rubs (50 passes over the surface) when tested in accordance with 4.6.22. Rubbing through to primer indicates a failure.

3.6.17 Cleanability.

The coating system shall maintain a minimum cleaning efficiency of not less than 75 percent when cleaned with system cleaner qualified to MIL-PRF-87937, Type IV, or MIL-PRF-85570 Type II, in accordance with 4.6.23.

3.6.18 Strippability.

A minimum of 90 percent of the coating system shall be stripped from the test panel within 24 hours and not more than four applications of chemical stripper or strippers when tested in accordance with 4.6.24. A chemical stripper may be specified to strip the topcoat and a different chemical stripper may be specified to strip the primer. However, no more than the aggregate four applications shall be applied to meet the above requirement.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality system of manufacturer.

Before submitting a product for qualification to this specification, the manufacturer shall have a quality system in place which holds a third party accreditation to AS 9100.

4.2 Qualification.

All products sold to this specification shall be listed, or approved for listing, on the associated qualified products list, QPL-32239. To qualify, the product shall meet all of the tests in this specification. A subset of testing shall be performed on the coating samples that have been subjected to the requirements of paragraphs 3.4.2, 3.4.3, 3.4.4, and 3.5.3. The subset of tests shall be viscosity, surface appearance, gloss, wet tape, GE impact, and MIL-PRF-83282 fluid resistance. All qualification testing is to be carried out by AFRL/MLSSO through the Cooperative Research and Development Agreement (CRDA) with University of Dayton Research Institute (UDRI). Suppliers shall provide their manufacturing/quality control specifications and test procedures for each system component at the time of qualification. All systems submitted for qualification or purchased as qualified products under this specification shall meet all manufacturer production and quality assurance specifications.

4.2.1 Qualification approval.

Coating systems supplied to this specification shall be listed, or approved for listing, on the Qualified Products List, QPL-32239. The activity responsible for maintaining the QPL is AFRL/MLSSO, 2179 12th Street, WPAFB, OH, 45433-7718, phone (937) 255-0945, fax (937) 255-0954. Information pertaining to qualification of coating systems may be obtained from this activity.

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4.2.2 Certification.

Qualified products shall be certified every two years to verify that there have been no changes in the material ingredients, manufacturing processes, or site of production since qualification and that the product meets all of the requirements of this specification (see 6.3.4).

4.2.2.1 Approval of minor changes.

The manufacturer shall use ingredients, manufacturing procedures, processes, location, and methods of inspection on production products which are the same as those used on the approval sample. If it is necessary to change ingredients, type of equipment for processing, or manufacturing procedures or location, the qualifying activity will request that the manufacturer submit for approval a statement of the proposed changes in ingredients and/or processing and, at the option of the qualifying activity, a sample product.

The qualifying activity, at its discretion, may accomplish partial or full requalification testing. Products made by the revised procedure shall not be shipped as a MIL-PRF-32239 qualified product prior to approval.

4.3 Quality assurance testing.

In order to ensure conformity to the composition and performance established during qualification, the manufacturer shall test a representative sample, taken in accordance with ISO 15528, of each batch of the material before shipment. As a minimum, conformance tests shall consist of the following:

a. 3.5.1 VOC content
b. 3.5.3 Viscosity and pot-life
c. 3.5.4 Drying time
d. 3.6.2 Color
e. 3.6.4 Gloss

Note: The quality assurance authority may take samples of the material and its ingredients at any stage of manufacture or from any portion of the consignment for inspection.

4.4 Standard conditions.

4.4.1 Test conditions.

Standard laboratory conditions shall be 77±5 ºF (25±3 ºC) and 50±10 percent relative humidity. Except as otherwise specified herein, all test specimens shall be cured under these conditions.

4.4.2 Standard tolerances.

Unless otherwise specified herein, Table III shows standard tolerances applied throughout test methods.

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TABLE III. Standard tolerances.

Measurement Units	Tolerances
Temperature	±2 ºF (±1 ºC)
Days	±2 hours
Hours	±5 minutes
Minutes	±10 seconds
Inches (mm)	±0.010 inch (0.25 mm)

4.5 Preparation of test panels.

a. Description of Test Panels.

Except as specified in the test procedures of 4.6 and as specified in Table IV, all test panels shall be 0.032 X 3 X 6 inch (.81 X 76.2 X 152.4 mm) bare aluminum alloy conforming to AMS-QQ-A-250/4. Test procedures 4.6.13 and 4.6.14 shall use an additional substrate conforming to AMS-4911. Test procedure 4.6.19 shall use substrates conforming to AMS-QQ-A-250/5 and AMS-QQ-A-250/13. Test Procedure 4.6.20 shall use additional substrate conforming AMS-QQ-A-250/12.

TABLE IV. Test panel plan.

Requirement Paragraph	Property	Alloy and Heat Treatment	Quantity of Panels Required
3.4.2 Storage Stability	Viscosity and Pot Life	N/A	N/A
	Surface Appearance	Bare 2024-T3	3
	Gloss	Bare 2024-T3	3
	Wet Tape	Bare 2024-T3	3
	Impact Flexibility Ambient	Bare 2024-O	3
	Fluid Resistance MIL-PRF-83282	Bare 2024-T3	3
3.4.3 Accelerated Storage Stability	Viscosity and Pot Life	N/A	N/A
	Surface Appearance	Bare 2024-T3	3
	Gloss	Bare 2024-T3	3
	Wet Tape	Bare 2024-T3	3
	Impact Flexibility Ambient	Bare 2024-O	3
	Fluid Resistance MIL-PRF-83282	Bare 2024-T3	3

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TABLE IV. Test panel plan – continued.

3.4.4 Freeze/Thaw Stability	Viscosity and Pot Life	N/A	N/A
	Surface Appearance	Bare 2024-T3	3
	Gloss	Bare 2024-T3	3
	Wet Tape	Bare 2024-T3	3
	Impact Flexibility Ambient	Bare 2024-O	3
	Fluid Resistance MIL-PRF-83282	Bare 2024-T3	3
3.5.2 Odor	Residual Odor of Dry Coating after 48-hrs	Bare 2024-T3	Use Panels from 3.6.1
3.5.3 Viscosity and Pot Life	Surface Appearance	Bare 2024-T3	3
	Gloss	Bare 2024-T3	3
	Wet Tape	Bare 2024-T3	3
	Impact Flexibility Ambient	Bare 2024-O	3
	Fluid Resistance MIL-PRF-83282	Bare 2024-T3	3
3.5.4 Dry Time	Dry-to-Tape	Bare 2024-T3 0.032 X 12 X 12 (0.813 X 304.8 X 304.8 mm)	1
3.5.5 Wet Edge	Wet Edge	Bare 2024-T3 0.032 X 12 X 12 (0.813 X 304.8 X 304.8 mm)	1
3.6.1 Surface Appearance	Surface Appearance	Bare 2024-T3	3
3.6.2 Color	Color	Bare 2024-T3	3
3.6.3 Infrared Reflectance	Visible Light, Ultraviolet, and Near Infrared Reflectance	Bare 2024-T3	3
3.6.4 Gloss	Gloss	Bare 2024-T3	Use Panels from 3.6.2
3.6.5 Opacity	Hiding Power	LENETA form 3B opacity card	

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TABLE IV. Test panel plan – continued.

3.6.6.1 Wet Tape	Adhesion (Wet Tape)	Bare 2024-T3	3
		Ti 6Al-4V	3
3.6.6.2 Cross Hatch	Adhesion (Cross Hatch)	Bare 2024-T3	3
		Ti 6Al-4V	3
3.6.7.1 Ambient Temperature Impact Flexibility	GE Impact Elongation (Ambient Temp)	Bare 2024-O .020 in (.51 mm) thick	3
3.6.7.2 Low Temperature Flexibility	Mandrel Bend Flexibility (Low Temp)	Bare 2024-O .020 in (.51 mm) thick	3
3.6.8 Fluid resistance	MIL-PRF-23699	Bare 2024-T3	3
	MIL-PRF-83282	Bare 2024-T3	3
	Skydrol LD-4	Bare 2024-T3	6
	JP-8 +100	Bare 2024-T3	3
	DI Water	Bare 2024-T3	3
3.6.9 Weathering (Xenon Arc)	Color	Bare 2024-T3	3
	Gloss	Bare 2024-T3	Use panels from color
	GE Impact Elongation (Ambient Temp)	Bare 2024-O .020 in (.51 mm) thick	3
	Mandrel Bend Flexibility (Low Temp)	Bare 2024-O .020 in (.51 mm) thick	3
3.6.9 Weathering (QUV-B)	Color	Bare 2024-T3	3
	Gloss	Bare 2024-T3	Use panels from color
	GE Impact Elongation (Ambient Temp)	Bare 2024-O .020 in (.51 mm) thick	3
	Mandrel Bend Flexibility (Low Temp)	Bare 2024-O .020 in (.51 mm) thick	3
3.6.10 Humidity Resistance	Blistering/Film Defects	Bare 2024-T3	3
	Cross Hatch Adhesion	Bare 2024-T3	Use panels from blistering
	Pencil Hardness	Bare 2024-T3	Use panels from blistering
3.6.11 Filiform Corrosion Resistance	Filiform Growth	Clad 2024-T3	3
		Clad 7075-T6	3
	Filiform Control	Clad 2024-T3	3

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TABLE IV. Test panel plan – continued.

3.6.12 Neutral Salt Spray Corrosion Resistance	Corrosion Evaluation	Bare 2024-T3	3
		Bare 7075-T6	3
3.6.13 Heat Resistance	Color	Bare 2024-T3	3
	Cross Hatch Adhesion	Bare 2024-T3	Use panels from color
	Mandrel Bend Flexibility	Bare 2024-O	3
3.6.14 Solvent Resistance (Cure)	MEK Rubs	Bare 2024-T3	3
3.6.15 Cleanability	Cleanability	Bare 2024-T3 0.063 X 3 X 6 inch (1.6 X 76.2 X 152.4 mm)	3
3.6.16 Strippability	Strippability	Bare 2024-T3	3

b. Test Panel Preparation.

Standard Panel Preparation. The standard panel preparation procedures in paragraphs 4.5.1, 4.5.2, and 4.5.3 are based upon practices long accepted by the aerospace community involving epoxy or polyurethane primers and topcoats.

c. Deviations from Standard Panel Preparation.

In order to encourage innovation and creativity, manufacturers may deviate from the standard panel preparation procedures (4.5.2). Manufacturers may utilize conditions, procedures, or techniques specific and unique to their materials and processes when attempting to qualify advanced concepts in aerospace finishing systems. (See 6.7 for guidance)

4.5.1 Control panel preparation.

Prepare the control panels for Filiform testing according to the following procedure:

a. Solvent wipe panels with methyl ethyl ketone.

b. Test panels shall be hand-scrubbed with an abrasive pad (Scotchbrite 7447B (red) pad) and a 10 percent by volume Brulin 815GD and 90 percent by volume DI water cleaning solution to remove surface contaminants and to lightly abrade the surface. Rinse the test panels in tap water.

c. Using a deoxidizing solution consisting of the following:

(1) 35 percent by volume n-butyl alcohol

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(2) 25 percent by volume isopropyl alcohol

(3) 22 percent by volume DI water

(4) 18 percent by volume ortho-phosphoric acid (85 percent)

Note: This formulation is derived from Method 2013.1 of FED-STD-141.

Immerse panel for 2 minutes and rinse with tap water and then with de-ionized or distilled water. Test for a water-break-free surface using distilled or de-ionized water. If the panel fails the water break test, repeat steps 2 and 3. Do not allow panel surfaces to dry between steps 3 and 4.

d. Apply a wash primer (MIL-C-8514) in accordance with MIL-C-8507.

e. Apply a gloss white, FED-STD-595, 17925, topcoat conforming to MIL-PRF-85285, TYPE I. Dry film thickness of the topcoat shall be 2.2±0.2 mils.

f. Control panels shall be cured 14 days at standard conditions.

4.5.2 Test panel preparation.

Alternate procedures for application of coating system components may be followed as specified by the manufacturer. Immersion applications shall not be used for the qualification of coating systems. In the absence of specific instructions to the contrary, prepare test panels according to the following procedure:

NOTE: If a non-chrome primer with a topcoat is submitted for qualification without a specified non-chrome pretreatment, they will be tested with an Alodine 1200S pretreatment and shall be listed as Type I (coating system containing chromium compounds) if the coating system passes all the requirements of this specification.

a. Solvent wipe panels with methyl ethyl ketone to remove ink stampings completely.

b. Test panels shall be hand-scrubbed with an abrasive pad (Scotchbrite 7447B (red) pad) and a 10 percent by volume Brulin 815GD and 90 percent by volume DI water cleaning solution to remove surface contaminants and to abrade the surface lightly. Rinse the test panels in tap water.

c. Using a deoxidizing solution consisting of the following:

(1) 35 percent by volume n-butyl alcohol

(2) 25 percent by volume isopropyl alcohol

(3) 22 percent by volume DI water

(4) 18 percent by volume ortho-phosphoric acid (85 percent)

Note: This formulation is derived from Method 2013.1 of FED-STD-141.

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Immerse panel for 2 minutes, rinse with tap water and then with deionized or distilled water. Test for a water break free surface using distilled or deionized water. If the panel fails the water break test, repeat steps 2 and 3. Do not allow panel surfaces to dry between steps 3 and 4.

d. Spray or brush the wet panel with Alodine 1200S according to manufacturers instructions. For test procedure described in paragraph 4.6.15.2, this step shall be omitted.

e. Apply primer(s) and topcoat(s) according to manufacturers' instructions. The thickness of each system component shall be maintained within the range specified by the manufacturer. Total dry film thickness of the complete coating system shall be less than 3.2 mils unless previously approved by AFRL/MLSSO.

f. All test panels shall be cured 14 days at standard conditions.

4.5.3 Test panel scribe.

Test panels that require a scribe shall be machined to the dimensions shown in figure 1 utilizing a New Hermes engraving machine and rotating carbide cutter as shown. Test panels for 4.6.16 Fluid Resistance, Skydrol test panels ONLY, shall be scribed with a 4±0.125 inch (101.6±3.175 mm) diagonal line. Test panels for 4.6.19 Filiform Corrosion and 4.6.19 Neutral Salt Spray corrosion testing shall be scribed with two 4±0.125 inch (101.6±3.175 mm) diagonal lines forming an "X" pattern in the center of the test panel. The diagonal lines shall be positioned equal distance and in line with the opposing corners of the test panel.

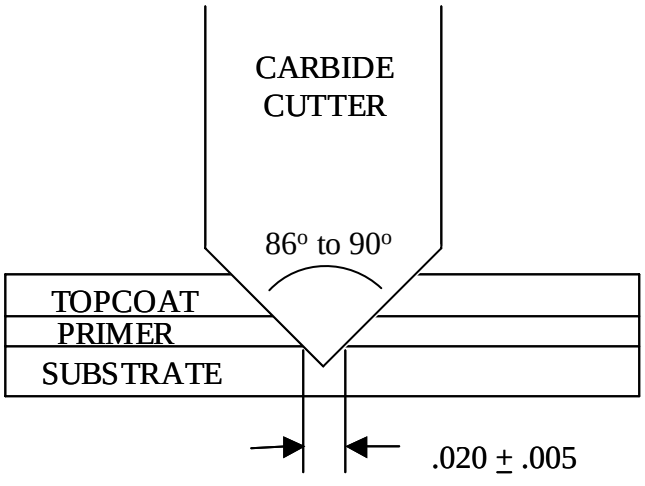


FIGURE 1. Dimensions for scribing test panels.

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4.6 Test procedures.

4.6.1 Storage stability.

Original unopened containers of primer, topcoat, or unicoat shall be stored indoors for one year from date of manufacture where the daily temperature of the ambient air is maintained between 35 to 115 ºF (1.7 to 46 ºC). All surface pretreatment concentrated liquids and powders shall be stored under the same conditions for 6 months.

Test panels for properties testing under this paragraph shall be made from aged system components. Test panels shall be created after the four hour pot-life has elapsed for the primer and topcoat.

4.6.2 Accelerated storage stability.

Each coating component, as packaged by the manufacturer shall be stored at 140±5 ºF (60±3 ºC) for 7 days, cool to room temperature and examine for conformance to 3.4.3.

Test panels for properties testing under this paragraph shall be made from accelerated aged topcoat applied over the accelerated aged primer. Test panels shall be created after the four hour pot-life has elapsed for the primer and topcoat.

4.6.3 Freeze/thaw stability.

The test shall be five cycles of freezing and thawing when tested in accordance with ASTM D 2243. One freeze/thaw cycle shall be 16 hours at 16 ±5 ºF (-9±3 ºC) followed by eight hours at room temperature 64 to 84 ºF (18 to 29 ºC).

Test panels shall be created after the four hour pot-life has elapsed for the primer and topcoat.

4.6.4 Volatile organic compound (V.O.C.) content.

The test shall be conducted in accordance with ASTM D 3960.

4.6.5 Odor.

After drying for 48 hours at laboratory conditions, place one panel into a 1000 mL tall form clean-dry glass beaker cover with a watch glass and allow to equilibrate for 20 minutes at laboratory conditions, then remove watch glass, raise the beaker to the nose and examine for the presence of an odor. Before placing the panel into the beaker, verify that there are no residual odors from the cleaning materials used to clean the beaker.

4.6.6 Viscosity/pot-life.

Where applicable, the test shall be conducted in accordance with ASTM D1200 using a #4 Ford viscosity cup. For pot-life, the admixed coating system components shall remain in a closed container and tested for viscosity in accordance with ASTM D 1200 using a #4 Ford viscosity cup four hours after mixing.

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Test panels for properties testing under this paragraph shall be made from 4-hour aged topcoat applied over the 4-hour aged primer.

4.6.7 Dry to tape.

After a double lap of tape is removed from the roll, a 5-inch strip of masking tape (3M #250), 1-inch (25.4 mm) wide, shall be applied to each test panel and pressed down with four double passes of a roller weighing no less than 4.5 lb (2.04 kg). The tape shall remain in contact with the test panel for no less than one hour. The tape shall be removed by seizing the free end and pulling it with a steady moderate pull, back on itself as close to 180º as possible.

4.6.8 Wet-edge.

The following test shall be conducted in a spray booth with an air flow of 75 to 125 linear feet per minute (0.381 to 0.635 meters per second). Use a 12 by 12 inch (304.8 by 304.8 mm) panel. Spray a coating layer of the mixed topcoat to half of the panel. After 15 minutes in the spray booth, spray a coating layer of the mixed topcoat over the second half of the panel overlapping the first application by one inch. After the second application, the panel may be removed from spray booth for final cure. Evaluation of test panel shall be after a minimum of 24 hour cure.

4.6.9 Color.

The test shall be conducted in accordance with ASTM D 2244 using CIE 1976 L* a* b*. Ensure that the spectrometer is set to the following parameters:

a. CIE Lab color system
b. 10º observer
c. D65 illuminant
d. Specular reflectance included

4.6.10 Infrared reflectance.

The total infrared reflectance (specular and diffuse) of the coating shall be measured using NIST traceable Spectralon white calibration standard over the wavelength range of 700 to 2600 nanometers (nm) with a Perkin-Elmer LAMBDA 9 spectrophotometer equipped with a total integrating sphere or equivalent.

4.6.11 Gloss.

The test shall be conducted in accordance with ASTM D 523.

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4.6.12 Opacity (hiding power).

The system topcoat shall be cast to a dry film thickness of 0.002 to 0.0024 inches (0.051 to 0.061 mm) onto three LENETA Form 3B Opacity cards, or equivalents, and tested in accordance with ASTM D 2805. The average opacity will be assessed for compliance.

4.6.13 Wet tape test.

Loss of adhesion as a result of exposure tests shall be determined in accordance with Method 6301.3 of FED-STD-141 with the following exceptions:

a. The test panels shall be thoroughly dried by wiping with clean dry cloth conforming to AMS 3819B, Grade A.

b. The test shall be conducted after no more than 10 minutes after wiping and standing at 77±5 ºF (25±3 ºC).

c. With a razor blade or scalpel, cut two parallel lines 1 inch (25.4 mm) apart and an X in between and overlapping the parallel lines. The X shall be 0.5 inches (12.7 mm) wide as shown in figure 2.

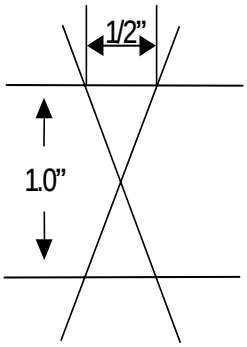


FIGURE 2. Wet tape test.

Parallel lines and an X shall be cut through the coating to bare metal to the dimensions shown using a razor blade or scalpel.

d. Press a piece of 1-inch (25.4 mm) wide masking tape (3M 250 tape) perpendicular across the parallel cuts covering the X. The tape shall be pressed down by using eight passes of a 4.5 to 5-pound (2.04 to 2.27 kg) rubber covered roller with a durometer hardness of Shore D 70-80, that is approximately 3.5 inches (88.9 mm) in diameter by 1.75 inches (44.45 mm) long.

e. Within 90 ± 30 sec. of rolling the tape, the tape shall be removed in one abrupt motion, pulling at an angle of 180º to the plane of the panel.

f. The adhesion is determined by a visual examination of the paint remaining/removed from the scribed area, using the rating scale in ASTM D 3359, Method A.

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4.6.14 Cross hatch.

The test shall be conducted in accordance with ASTM D 3359, Method B using a Gardco cutting tool (part number PA-2058 or PA-30XX). The blade used to make the cross hatch shall make 6 cuts that are 2 mm apart.

4.6.15 Flexibility.

4.6.15.1 Ambient temperature impact flexibility.

The coating system shall be tested in accordance with ASTM D 2794, using the Gardco GE universal impact tester, model 172 or equivalent. Cracking of the coating is defined as cracking to the substrate and will be assessed using an Elcometer, Model D204-01 holiday detector or equivalent.

4.6.15.2 Low temperature flexibility.

Class 1 coating systems:

The coating system shall be tested in accordance with ASTM D 522 Method B at -60±5 ºF (-51±3 ºC) using a 1-inch (25.4 mm) mandrel for gloss and semi-gloss and a 2-inch (50.8 mm) mandrel for camouflage. Class 2 coating systems:

The coating system shall be tested in accordance with ASTM D 522 Method B at -60±5 ºF (-51±3 ºC) using a 1” (25.4 mm) mandrel for gloss, semi-gloss and camouflage topcoat..

4.6.16 Fluid resistance.

Use test panels prepared in accordance with 4.5 through 4.5.2. To establish the initial values for pencil hardness, use 2024-T3 test panels from 3.6.6.2 in accordance with 4.6.25. The initial adhesion values will be taken from these same test panels. Expose a separate set of three test panels to each of the following fluids at the designated conditions:

a. MIL-PRF-23699 – Prior to exposure, measure the color of three panels in accordance with 4.6.9, and then completely immerse them for 24 hrs @ 250±5 ºF (121±3 ºC).

b. MIL-PRF-83282 – Prior to exposure, measure the color of three panels in accordance with 4.6.9, and then completely immerse them for 7 days @ 150±5 ºF (65.6±3 ºC)

c. Jet Fuel JP-8 +100 – Prior to exposure, measure the color of three panels in accordance with 4.6.9, and then completely immerse them for 30 days @ 77±5 ºF (25±3 ºC)

d. Water (ASTM D 1193 Type IV) – Prior to exposure, measure the color of three panels in accordance with 4.6.9, and then completely immerse them for 30 days @ 120±5 ºF (49±3 ºC)

e. Skydrol LD-4 – Prior to exposure, measure the color of three panels in accordance with 4.6.9, then scribe three additional panels in accordance with 4.5.3, and then place the six panels in a horizontal position coating side up in a test area maintained at 77± 5 ºF (25±3 ºC). Using Skydrol LD-4 at 77±5 ºF (25±3 ºC), wet the six panels once each day (no immersion) for 30 days.

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After each exposure, remove the panels, and immediately evaluate for blistering and film delamination. After evaluation, gently clean the MIL-PRF-23699, MIL-PRF-83282, Jet Fuel, and the unscribed Skydrol LD-4 panels with the coating system cleaner, and rinse thoroughly with water. Blot the panels dry and allow to air dry for 1-hour, and then measure the final color in accordance with 4.6.9. After the color measurement has been made, test for the final adhesion and hardness in accordance with 4.6.14, and 4.6.25 respectively. Compare the average of the initial values with the average of the final values to assess compliance.

4.6.17 Weather resistance.

4.6.17.1 Xenon arc.

Test panels shall be exposed for 3000 Hrs. in a Xenon-arc weather-o-meter (Q-Sun/3000 or Q-Sun/1000) that is cycling between 102 minutes of light only and 18 minutes of light and DI water spray. After exposure, examine for conformance to 3.6.9. The following conditions shall apply when tested in accordance with ASTM G 155, Type BH:

Black body temperature in cabinet:	145±5 ºF (63±3 ºC)
Intensity (spectral irradiance) of Xenon-arc:	0.35 ± 0.05 watt/meter^2 at a wavelength of 340 nm

4.6.17.2 QUV-B.

Test panels shall be tested in accordance with ASTM G 154 using an exposure of 1500 Hrs. in a UV-CON cabinet that is cycling between 8 hours with a UVB-313 lamp at 140 ºF (60 ºC) followed by 4 hours of darkness and water condensation at 113 ºF (45 ºC). The UV-CON cabinet shall have the control wavelength set at 310 nm and the irradiance set at 0.63 watts/m^2.

4.6.18 Humidity resistance.

Test panels shall withstand exposure for no less than 30 days in a humidity cabinet maintained at 120±3 ºF (48.9±1.6 ºC) and 100 percent relative humidity in accordance with ASTM D 2247.

4.6.19 Filiform corrosion.

Test and control panels shall be prepared in accordance with 4.5 through 4.5.2 and scribed in accordance with 4.5.3. The panels shall be supported 2 inches above the liquid level in a closed container containing sufficient concentrated hydrochloric acid solution to completely cover the bottom of the container for 65±5 minutes. Do not rinse or dry. Immediately transfer the panels directly to a humidity chamber maintained at 102±5 ºF (38.9±3 ºC) and 80 percent ±5 percent relative humidity. Support the panels with the long dimension 6º from the vertical and the short dimension horizontal. The test duration shall be a minimum of 2000 hours. After exposure is complete and in preparation for panel evaluation, chemically strip the coating system from half of the panel from the short dimension.

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4.6.20 Neutral salt spray corrosion.

Type 1: Test panels prepared in accordance with 4.5 through 4.5.2 and scribed in accordance with 4.5.3 shall be exposed for 3000 hours in a salt spray cabinet that is using a 5 percent solution in accordance with ASTM B 117.

Type 2: Test panels prepared in accordance with 4.5 through 4.5.2 and scribed in accordance with 4.5.3 shall be exposed for 2000 hours in a salt spray cabinet that is using a 5 percent solution in accordance with ASTM B 117.

4.6.21 Heat resistance.

Test panels shall be placed in a convection oven at 350±5 ºF (176.6±15 ºC). After four hours, remove the panels from the oven and allow them to cool to standard conditions. The –O temper test panels shall be tested for flexibility in accordance with ASTM D 522, method B using a 1-inch (25.4-mm) mandrel for gloss coatings, and a 2-inch (50.8-mm) mandrel for camouflage coatings. The -T3 test panels shall be tested for adhesion in accordance with 4.6.14.

4.6.22 Solvent resistance (cure).

Test panels shall be prepared in accordance with 4.5 through 4.5.2. A cotton, terrycloth rag shall be soaked in methyl ethyl ketone (MEK) solvent and rubbed back and forth 25 times (50 passes) over the coating with firm finger pressure.

4.6.23 Cleaning efficiency.

The cleaning efficiency test shall be performed in accordance with 4.6.22.1 through 4.6.22.7.

4.6.23.1 Preparation of artificial soil.

Prepare the artificial soil by mixing 1.76 ± 0.018 oz (50.0±0.5 g) of carbon black with 17.64±0.035 oz (500.0±1.0 g) of lubricating oil conforming to MIL-PRF-23699. Homogenize the artificial soil using a high shear mixer for 15±1 minutes. Prior to each application of the artificial soil, stir or shake the mixture by hand.

4.6.23.2 Cleaning compound.

Mix MIL-PRF-87937, Type IV or MIL-PRF-85570, Type II cleaning compounds in accordance with the manufacturer's instructions.

4.6.23.3 Preparation of test panels.

Prepare two test panels in accordance with 4.5 through 4.5.2. Using a clean, hog bristle brush, lightly scrub the coating of each panel with 4.6.24.2 cleaning compound. Rinse each panel thoroughly three times in reagent water. Dry for 18 hours at 120±5 ºF (48.9±3 ºC). Using ASTM D 2244, determine the L* value of the coating. This will be value "A".

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4.6.23.4 Soiling of test panels.

Using a soft-bristle brush, coat the painted surfaces of the test panels with the artificial soil. Remove excess soil by covering the test panel surface with folded absorbent tissue and exerting pressure by rolling the tissue with a 5 lb (2.27 kg) rubber roller. Repeat this blotting procedure twice. Brush the soiled surface only in one direction and parallel to the long dimension of the test panel, using 10 strokes of the hog bristle brush. Bake the test panel at 221±5 ºF (105±3 ºC) for 60±1 minutes. Determine the L* value of the coating system in accordance with ASTM D 2244. This will be value “B.” For the test to be valid, “B” shall be 25±5. Test panels outside this range must be discarded and new test panels created.

4.6.23.5 Procedure.

Clean the soiled test panel within 4 hours of soil application using a Gardner Heavy Duty Wear Tester, Model D10V, or equivalent, with a sponge attached to the wear head.

4.6.23.6 Cleaning.

Using hook-and-loop (Velcro) type strips, attach the nylon web side of a dry sponge, dimensions 3.5 by 2.75 inch (88.9 by 69.85 mm) to the cleaning head of the wear tester, such that the dimension parallel to the cleaning stroke is 3.5 inch (88.9 mm). The combined weight of the sponge and cleaning head shall be 2.98 to 3.09 lbs (1350 to 1400 grams). Place the test panel holder in the wear tester. Place a soiled test panel in the holder such that the test panel is at +45º to the cleaning stroke. Saturate the sponge and cover the test panel with the 4.6.24.2 cleaning compound. After 60±5 seconds, clean the test panel using 5 cycles of the wear tester, then immediately turn the test panel 90º in the template and clean for an additional 5 cycles. Remove the test panel from the holder and rinse the test panel with a flowing stream of water at room temperature. Allow the test panel to dry. Using ASTM D 2244, determine the L* value of the coating system. This will be value “C”.

4.6.23.7 Calculation.

Cleaning efficiency (percent) = [(C-B)/(A-B)] X 100

4.6.24 Strippability.

Test panels shall cure under standard conditions for 7 days followed by an artificial aging at an elevated temperature of 210 ºF for 96 continuous hours. All test panel edges shall be masked with 3-M aluminum tape and placed on a rack at a 60º angle from horizontal. Enough chemical stripper qualified to one of the chemical stripper documents listed below, shall be poured along the upper edge of each test panel to completely cover the coating surface. After 6 hours of exposure, the loosened coating system shall be brushed off, and the test panel shall be rinsed while brushing under a stream of cool water. Successive applications of the same chemical stripper or different stripper may be applied in the same manner. No more than 4 applications of strippers totaling 24 hours shall be applied. The amount of coating stripped in this test is determined by the percentage of substrate surface area exposed.

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AUTHORIZED CHEMICAL STRIPPER PRODUCTS:

Warner-Robins ALC Purchase Description (U.S. Air Force)

Oklahoma City ALC Purchase Description (U.S. Air Force)

Technical Order 1-1-8 (U.S. Air Force)

4.6.25 Pencil hardness.

Test panels shall be placed in a horizontal position and tested immediately at standard conditions in accordance with ASTM D 3363 with the exception that the desired edge of the pencil lead shall be maintained before each stroke.

4.7 Reports.

The supplier of the product shall furnish with each shipment a certificate of conformity. This certificate shall include the purchase order number, lot number, MIL-PRF-32239, manufacturer's identification, and quantity.

4.7.1 Resampling and retesting.

If any specimen used in the above tests fails to meet the specified requirements, disposition of the product may be based on the results of testing three additional specimens for each nonconforming specimen. Failure of any retest specimen to meet the specified requirements shall be cause for rejection of the product represented. Results of all tests shall be reported.

5. PACKAGING

5.1 Packaging.

For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use.

The coating/finish system covered by this specification is intended for use on external moldlines of military aerospace weapon systems. In general, the components comprising the coating system work in concert to provide superior weather and chemical resistance with protection against corrosion and may be used in applications other than aerospace weapon systems. Additional testing would be needed to address the specific requirements of other applications. This specification contains military specific camouflage gloss requirements, gunship gloss requirements, and low IR requirements.

6.2 Acquisition requirements.

Acquisition documents should specify the following:

a. Title, number and date of this specification.

b. Type, class and grade of the coating system desired.

c. Quantity and size of containers for each system component required.

d. Level of packaging and packing (see 5.1 and 6.10)

e. Minor modifications desired to one or more components in the coating system relative to unique application conditions, such as faster cure.

f. Information required on toxicity, material safety data and precautionary measures (see 3.3.2 and 6.3.3).

6.3 Qualification.

At the time set for opening of bids, awards will be made only for coating systems which are listed on the qualified products list of this specification. The attention of suppliers is called to this requirement; manufacturers should arrange to have their proposed coating system tested for qualification prior to being eligible for contract awards or orders.

6.3.1 Qualification of gloss or camouflage coating systems.

Gloss and camouflage coating systems will be qualified separately. Manufacturers may qualify to either or both types. Upon qualification it is accepted that all FED-STD-595 colors in the same gloss category are qualified. If both gloss and camouflage coating systems pass qualification, it is accepted that semi-gloss coatings are qualified. To qualify a gloss coating system the manufacturer should submit FED-STD-595 color 17925. To qualify to camouflage coating system the manufacturer should submit FED-STD-595 color 36173. In addition, for the purpose of opacity testing, FED-STD-595 colors 13538, 12197, and 11136 should be submitted.

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6.3.2 Qualification inspection samples.

The test samples should consist of a minimum of three one-quart kits and one one-gallon kit for the primer, and topcoats. For opacity testing the samples should consist of minimum one-pint kits. The material should be furnished in the type of containers to be used in filling contract orders. Samples should be identified as follows and forwarded to the laboratory designated in the letter of authorization.

6.3.3 Material Safety Data Sheets (MSDS).

Contracting officers should identify those activities requiring copies of MSDS's prepared IAW FED-STD-313. The pertinent Government mailing addresses for submission of data are listed in FED-STD-313; and 29 CFR 1910.1200 requires that the MSDS for each hazardous chemical used in an operation must be readily available to personnel using the material. Contracting officers should identify the activities requiring copies of the MSDS.

6.3.4 Certification of qualified products.

A request for certification consists of a letter or form DD 1718 certifying that there have been no changes in the material ingredients, manufacturing processes, or site of production since qualification and that the product meets all of the requirements of this specification. In addition a test report should be provided from a laboratory of the manufacturer's choosing, showing compliance with all of the qualification tests. AFRL/MLSSO may elect to certify without testing based on this report, or may require partial or full requalification testing before the manufacturer may ship a product with the MIL-PRF-32239 label.

6.4 Government laboratory.

References throughout this specification to a government or testing laboratory is the Air Force Research Laboratory, Materials and Manufacturing Directorate, attn: AFRL/MLSSO (CTIO), 2179 12th Street, WPAFB, Ohio, 45433-7718, voice (937) 255-0945, fax (937)255-0954.

6.5 Coating system definition.

The definition of a coating system qualified to this specification in 1.1 includes a chemical stripper and a cleaning compound. It is not the intent of this specification or of the above government testing laboratory to qualify chemical strippers and cleaning compounds. The manufacturer is required to select current qualified cleaning products (see 4.6.22.2) and chemical strippers (see 4.6.23) that give optimal performance to the coating system being submitted for qualification. A manufacturer may formulate the components of their coating system to achieve maximum cleanability and strippability with respect to specific qualified cleaners and strippers while meeting all requirements of this specification.

6.6 Manufacturer partnerships.

The formation of partnerships among individual component suppliers should result in superior overall coating system performance. However, partnerships between individual manufacturers of coating system components are not required to achieve listing on the qualified products list.

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6.7 Innovation.

While the concept of an advanced performance coating system is not new, the encouragement within this specification of innovation and use of emerging technologies to produce and qualify an advanced performance aerospace coating system is new. Deviations from current best aerospace practices, materials and processes are allowed but the resulting coating system must meet or exceed the performance requirements of this specification and be able to perform in the maintenance environment of legacy aerospace weapon systems. For guidance on depot and field level maintenance environments, contact the above government testing laboratory.

6.8 Special application conditions.

Procuring activities may request minor solvent and/or additive modifications to qualified products to solve unique application problems, such as faster drying time and shorter pot-life coatings formulated for plural-component spray equipment. Only application conditions are permitted to be modified. The cured coating (dry film) properties of the coating are not permitted to be changed. Any modified coatings must be clearly identified as such on its container and may only be furnished to the requesting activity. The specific modifications and labeling instructions should be detailed in any procurement request for the modified material.

6.9 Definitions.

Definitions of terms and expressions used for coatings as given in ISO 4617 and ISO 4618 apply, with the exception of those defined below:

a. Batch. Quantity of one component of a product with the same composition and resulting from the same production run.

b. Coating system. The materials and processes necessary for finishing an aircraft consisting of one or more applications of specific products, methods or techniques necessary to fulfill the requirements of this specification. This definition includes materials and processes for surface pretreatment, cleaning and chemical removal.

c. Livering. Related to gelling when a material takes on body caused by the evaporation of solvents or the oxidation of the material.

d. OEM. Original equipment manufacturer.

e. Pot-life. The time period after the coating is mixed for which it can be applied and meet the performance requirements of this specification.

f. Volatile organic compound (VOC). VOC is composed of one or more volatile organic compounds, containing at least elements of carbon and hydrogen or in which hydrogen is partially or completely replaced by halogens, oxygen, sulfur, phosphorus or nitrogen, and with the exception of carbon monoxide, carbon dioxide and carbonates, and which has a vapor pressure of 10 mmHg (1.3 kPa) or more, at 68 ºF (20 ºC). This specification recognizes by exception, the use of EPA defined exempt solvents.

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6.10 General packaging information.

The following packaging and packing procedures are applicable, wherever possible, to the various components of the coating system. Since the coating system concept does not lend itself to kit packaging, these requirements may be modified by the procurement document (see 6.2). At times, the procurement document may state the purchase of only one or two components of a coating system as they appear on the qualified products list.

6.10.1 Marking and labeling.

The container or package of each component or materials comprising a component of a qualified coating system should be marked in accordance with MIL-STD-129. In addition to the markings specified in MIL-STD-129, individual containers/packages should bear a printed, weather-proof label showing the following nomenclature and information as applicable.

Pretreatments:

- Title, number, and date of the specification
- Type required (see 1.3)
- Unit of issue required
- Quantity required
- Lot size
- Packaging requirements
- Warning and precautions as applicable

Primers:

- Title, number, and date of the specification
- Component Identification
  - Component A – Pigmented resin component
  - Component B – Catalyst (curing agent) resin component
  - Component C – Reducer
  - Component D – Single component
- Name and plant address of manufacturer
- Date of manufacture by month and year
- Type, class, and grade required (see 1.3)
- Lot number/net contents
- Storage instructions
- Warning and precautions as applicable

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Topcoats:

Title, number, and date of the specification

Component Identification

Component A – Pigmented resin component

Component B – Catalyst (curing agent) resin component

Component C – Reducer

Component D – Single component

Name and plant address of manufacturer

Date of manufacture by month and year

Class required (see 1.3)

Lot number/net contents

Storage instructions

Warning and precautions as applicable

6.11 Subject term (key word) listing.

chromium
chromium-free
flexibility
fluid resistance
low-IR
pigment
reflectance

Concluding Material

Custodians:
Army – MR
Navy – AS
Air Force – 11

Preparing Activity
Air Force – 11

(Project 8010-2006-025)

NOTE: The activities listed above were interested in this document as of the date on the cover. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://assist.daps.dla.mil.

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